Gate detection — what works & what is left to test

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TL;DR (revised after report 12)

Headline checkpoint revised. The previous report-07 ckpt (`best_deeplabv3plus_keypoint.pth`, single-gate target) was shown by report 12 to have memorised `flight-01p-ellipse`; on the only piloted flight held out from SAM training (`flight-06p-ellipse`, 0 train crops) it produces 0 % detection. The Stage 2 sibling (`best_deeplabv3plus_keypoint_multi.pth`, multi-gate Gaussian targets + `coherent_quad_decode`) generalises and is the new deployment checkpoint, delivering **78.9 % detection rate** at **1.30 px median / 61.0 % within 5 px** on the same held-out flight, single forward pass, no prior, no EKF state, **0.44 ms TRT FP16 / 2274 fps on an RTX 5090**. Upper-bound false-positive rate on no-gate frames of the held-out flight: **2.4 %** (4/166 at conf ≥ 0.3). What is *not* yet measured: bench on the autonomous held-out flights in SAM val (7 `XXa` flights), real-recorded footage (blocked on Drive data), bootstrap CI on the reported percentages.

1 What works today

Numbers are on `flight-06p-ellipse` (TII v3.0.0 piloted, **held-out — 0 SAM training crops**, 374 frames sampled at stride 30, seed 42), reproducible from report 12.

Segmentation backbone.
DeepLabV3+ with ResNet-34 + ImageNet pretrain — val IoU **0.872**. *report 05 / 07*

Keypoint head.
Multi-task 4-channel Gaussian-heatmap head, sub-pixel decode via soft-argmax. *report 07*

Deployment-mode bench (`model_direct`, held-out 06p).
78.9 % detection / **1.30 px median** / 61.0 % within 5 px on flight-06p-ellipse (no GT prior, no EKF state, 0 SAM training crops). *report 12*

gt-prior bench on the same held-out flight.
Confirms seg head transfers: **95.8 % within 5 px** (perfect-prior QuAdGate refinement) — the seg head learned generalisable features from the SAM-mask pipeline. *report 12*

False-positive rate (upper bound, held-out 06p, conf $\geq$ 0.3).
2.4 % (4/166 sampled no-gate frames). Upper bound because labellers skip non-trivial frames — a "fire" may be detecting an unlabelled gate. *report 12*

Fire-condition tightening.
`coherent_quad_decode is None` no longer falls through to soft-argmax; the predictor returns a non-detection cleanly (CLAUDE.md §6). Cost: 2/297 lost detections; structural FP win (no Frankenstein quads). *task 3, this turn*

Prior-mode framework.
Six priors wired in one bench script: `gt`, `noisy`, `none`, `model`, `model_direct`, `ekf`. *reports 06 / 07 / 08*

TRT FP16 export (multi-gate ckpt, the new headline).
Multi-task graph exports to ONNX (opset 18) and builds an FP16 TRT engine in ~ 7 s. **0.44 ms / 2274 fps on RTX 5090**, projects ∼50–100 fps on Jetson Orin NX. *`bench_trt_keypoint.py`*

TRT numerical correctness.
TRT FP16 output matches torch FP32 to $|trt - torch|_\infty < 1.3\times10^{-3}$ (seg) / 2.4×10^{-4} (kp) on uniform-$[0, 1]$ input. *same*

Segmentation backbone shared.
DeepLabV3+ with ResNet-34 + ImageNet pretrain — val IoU **0.872** on SAM-cropped val. NB: this is per-crop IoU on the SAM val pool (mostly 06p crops), not full-frame IoU on raw flight images. *report 05 / 07*

Knock-on consequences ruled in. The architecture exports cleanly to TRT; multi-gate-target training generalises across flights (report 12); the deployment headline is now on a truly held-out flight, not an over-trained one. The single-gate ckpt (`best_deeplabv3plus_keypoint.pth`) is retained as a sibling for any future flight-specific use case but is not the deployment ckpt.

2 What is *not* measured (and matters for detection reliability)

#	Open test	Cost
1	Bench remaining held-out flights (the 7 autonomous `XXa` flights in SAM val) on the multi-gate ckpt — gives an out-of-piloted held-out generalisation curve.	1 h
2	Bootstrap CI on all reported percentages (report 12's reviewer recommendation). Wilson CI on iid samples; cluster-bootstrap if temporal correlation matters.	1 h
3	Visually inspect the 12 multi-gate `01p-ellipse` "fires" on no-gate frames — separate "model triggered on unlabelled real gate" (not actually FP) from true precision loss.	30 min
4	Per-flight seg-IoU dump to attribute cross-flight degradation to seg vs head vs QuAdGate.	1 h
5	Confidence calibration: replace *peak* ≥ 0.3 with peak + spatial concentration (KL divergence of the peak's 7×7 window from a $\sigma = 3$ Gaussian). Catches the remaining FPs surviving `coherent_quad_decode`.	2 h
6	Retrain with class-balanced SAM crops (∼300/flight cap, removes 01p's 3× training advantage). Addresses the root cause behind report 12's findings.	1 day
7	Bench on `flight-01p-fixed` (the original 82 % headline): blocked on Drive data + missing `scripts/prepare_labels.py`.	blocked
8	Real-footage bench (the README's 39 % number): blocked on Drive data.	blocked
9	Real-data fine-tuning (encoder frozen, decoder + head on team-recorded footage).	blocked
10	Heavier motion-blur augmentation + retrain. Current aug caps at 15 px blur kernel.	1 day

What the open tests do *not* cover. "Which gate is the next one to fly through" is not a perception question under the current architecture — it is a planner-side concern that needs an EKF + track-plan + plane-test (the MonoRace recipe, Bahnam et al. 2026). That workstream is items 4–6 on the master rank list (Kalman fusion, ROS2, Orin NX) and intentionally outside the scope of "are gates detected reliably."

3 Recommended order

The original items 1–3 from this status note (cross-flight bench, FP-rate measurement, multi-gate sibling cross-flight) were done in report 12 and this turn's task batch. Headlines were revised accordingly. The revised next steps:

1. **Item 6 (class-balanced retrain)** addresses the *root cause* of report 12's findings: the SAM training pool over-represents 01p-ellipse 3×, which is why the single-gate ckpt memorised it. Cap per-flight at ∼300 crops and retrain both ckpts. ∼1 day.
2. **Item 1 (autonomous held-out bench)** — closes the held-out picture beyond just 06p. ∼1 h.
3. **Items 2–4 (CI + visual inspection + per-flight IoU)** — statistical rigour + attribution. ∼2–3 h combined.
4. **Item 5 (confidence calibration)** only if items 3–4 show the remaining 2.4 % FP rate is a real precision loss, not labeller-skipped real gates. ∼2 h.

Items 7–9 unblock once the Drive data lands on qolam; item 10 is the only remaining unblocked retrain lever and only justifies its cost if held-out flights show motion-blur-specific failures.

Bottom line. The model detects gates with **78.9 % recall / 1.30 px median / 2.4 % upper-bound FP rate** on the only piloted flight held out from SAM training, and we have enough plumbing (TRT export verified, prior-mode framework, two checkpoints) to ship. The single biggest remaining lever is the class-balanced retrain (item 6, root cause from report 12) — everything else is incremental.